

Q.1 In basis property the first iteration of the loop is solved for?

- A. None
 - B. $I(k)$
 - C. $I(k + 1)$
 - D. $I(0)$**
-

Q.2 The sum of a rational and irrational number

- A. none of these
 - B. may or may not be rational number
 - C. is always an irrational number**
 - D. is always a rational number
-

Q.3 In the proof by contradiction, we lead the assumption to -----

- A. None of these.
 - B. contingency
 - C. Absurdity**
 - D. tautology
-

Q.4 A predicate is a sentence that contains a finite number of -----

- A. Constants
 - B. variables**
-

Q.5 A non-zero integer d divides an integer n if and only if there exists an integer k such that -----

- A. $n = d k$**
 - B. $n = d / k$
 - C. $n = d - k$
 - D. $n = d + k$
-

Q.6 Algorithm is a more general term in that the term ---- refers to a particular programming language.

- A. Task
 - B. Program**
 - C. Information
 - D. None
-

Q.7 The predicate which describes the initial state is called the ----- of the algorithm

- A. post-condition
 - B. pre-condition**
-

Q.8 The sum of two integers is even when

- A. one is zero and one is odd
 - B. one is even and one is odd
 - C. one is prime and one is composite
 - D. both are even or both are odd**
-

Q.9 The sum of $(6 - 7\sqrt{2})$ and $(6 + 7\sqrt{2})$ is

- A. $2(6-7\sqrt{2})$
 - B. $12-14\sqrt{2}$
 - C. $2(6+7\sqrt{2})$
 - D. 12**
-

Q.10 For every prime number n , $n + 2$ is prime. statement.

Which of the following prime number disproves the above

- A. $n=11$
 - B. $n=5$
 - C. $n=7$**
 - D. $n=3$
-

Q.1 For integers a, b, c , if a divides b and a divides c , then a divides $(a + b)$.

- A. False
 - B. True**
-

Q.2 For every prime number n , $n + 2$ is prime.
statement.

Which of the following prime number disproves the above

- A. $n=5$
 - B. $n=11$
 - C. $n=7$**
 - D. $n=3$
-

Q.3 The quantity $a - d \cdot q$ equals the?

- A. None
 - B. Quotient
 - C. Divisor
 - D. Remainder**
-

Q.4 In division algorithm the pre-condition is?

- A. a is a nonnegative integer and d is a positive integer**
 - B. q and r are nonnegative integers
-

Q.5 The product of any two consecutive positive integers is divisible by 2.

- A. True**
 - B. False
-

Q.6 The last step of describing algorithms formally is?

- A. An end statement.**
 - B. The input variable names.
 - C. The name of the algorithm.
 - D. The output variable names.
-

Q.7 The sum of two integers is not an even integer when

- A. both are odd**
 - B. one is 0 and one is even
 - C. one is even and one is odd
 - D. both are even
-

Q.8 The sum of two irrational numbers in general

A. may or may not be an irrational number

B. none of these

C. is always a rational number

D. is always an irrational number

Q.9 If 'n' is an odd integer then n^3+n is

A. odd

B. even

Q.1 The method of loop invariants is based on the principle

of **A. Mathematical induction**

B. Division algorithm

Q.2 Which of the following statement is true according to the Division Algorithm?

A. $17 = 5 \times 3 + 2$

B. $17 = 5 \times 5 - 8$

C. $17 = 5 \times 4 - 3$

D. $17 = 5 \times 1 + 12$

Q.3 In the proof by contradiction, we lead the assumption to -----

A. tautology

B. Absurdity

C. None of these.

D. contingency

Q.4 A proof by contraposition is based on the logical equivalence between a statement and its.....

A. none of these

B. converse

C. contrapositive

D. inverse

Q.5 The sum of two integers is not an even integer when

A. both are even

B. both are odd

C. one is even and one is odd

D. one is 0 and one is even

Q.6 Proof byuses the equivalence $p \rightarrow q \equiv \sim q \rightarrow \sim p$.

A. mathematical induction

B. superposition

C. contradiction

D. contraposition

Q.7 The sum of a rational and irrational number

A. is always an irrational number

B. none of these

C. is always a rational number

D. may or may not be rational number

Q.8 For integers a, b, c , if a divides b and a divides c , then a divides $(a + b)$.

A. False

B. True

Q.9 The loop is correct if the four properties are true, one of the property is?

A. Transitivity

B. Associativity

C. Eventual Falsity of Guard

D. Commutativity

Q.10 While proving by contradiction the equivalenceis used.

A. $\neg p \rightarrow q \equiv (p \wedge \neg q) \rightarrow c$

B. $p \rightarrow \neg q \equiv (p \wedge \neg q) \rightarrow c$

C. $p \rightarrow q \equiv (p \wedge \neg q) \rightarrow c$

D. $p \rightarrow q \equiv (\neg p \wedge \neg q) \rightarrow c$

Q.1 The sum of two irrational numbers in general

A. may or may not be an irrational number

B. none of these

C. is always a rational number

D. is always an irrational number

Q.2 The loop is correct if the four properties are true, one of the property is?

A. Transitivity

B. Eventual Falsity of Guard

C. Associativity

D. Commutativity

Q.3 The contrapositive of "5n + 2 is odd, then n is odd" is

A. " if n is even then 5n + 2 is not even"

B. " if n is not even then 5n + 2 is even"

C. " if n is not even then 5n + 2 is not even"

D. " if n is even then 5n + 2 is even"

Q.4 Proof byuses the equivalence $p \rightarrow q \equiv \sim q \rightarrow \sim p$.

A. superposition

B. mathematical induction

C. contradiction

D. contraposition

Q.5 The sum of two integers is not an even integer when

A. one is 0 and one is even

B. one is even and one is odd

C. both are odd

D. both are even

Q.6 The sum of any rational number and any irrational number is irrational.

A. True

B. False

Q.7 The set of prime numbers is

A. infinite set.

B. None of these.

C. finite set.

D. continuous set.

Q.8 If 'n' is an odd integer then ' $3n+2$ ' is

- A. odd
 - B. even**
-

Q.9 If the square of an integer is even , then that integer is

- A. even**
 - B. neither even nor odd
 - C. either even or odd
 - D. odd
-

Q.10 The rational number '2' is the product of which of the two irrational numbers.

- A. $\sqrt{2}, \sqrt{3}$
 - B. $\sqrt{2}, \sqrt{7}$
 - C. $\sqrt{2}, \sqrt{2}$**
 - D. $\sqrt{2}, \sqrt{5}$
-

Q.1 The sum of two integers is not an even integer when

A. one is even and one is odd

B. both are odd

C. both are even

D. one is 0 and one is even

Q.2 For every prime number n , $n + 2$ is prime. statement.

Which of the following prime number disproves the above

A. $n=3$

B. $n=7$

C. $n=5$

D. $n=11$

Q.3 A proof by contraposition is based on the logical equivalence between a statement and its.....

A. none of these

B. inverse

C. contrapositive

D. converse

Q.4 An integer ' n ' is if it can be represented as a multiple of 2.

A. composite

B. prime

C. odd

D. even

Q.5 In inductive property the first iteration of the loop is solved for?

A. None

B. $I(k + 1)$

C. $I(0)$

D. $I(k)$

Q.6 For integers a, b, c , if a divides b and a divides c , then a divides $(a + b)$.

A. True

B. False

Q.7 Algorithm is a more general term in that the term ---- refers to a particular programming language.

A. Task

B. Program

C. Information

D. None

Q.8 The predicate describing the final state is called ____.

A. post-condition of the algorithm

B. Input variables

C. None

D. pre-condition of the algorithm

Q.9 In division algorithm the pre-condition is?

A. a is a nonnegative integer and d is a positive integer

B. q and r are nonnegative integers

Q.10 If the square of an integer is even , then that integer is

A. either even or odd

B. odd

C. neither even nor odd

D. even

Q.1 The sum of two integers is not an even integer when

- A. one is 0 and one is even
 - B. both are odd**
 - C. both are even
 - D. one is even and one is odd
-

Q.2 An integer 'n' is if it can be represented as a multiple of 2.

- A. even**
 - B. odd
 - C. prime
 - D. composite
-

Q.3 Which of the following is true for $n = 1$?

- A. $2^{(2n)} - 1$ by 5
 - B. $2^{(2n)} - 1$ by 2
 - C. $2^{(2n)} - 1$ by 4
 - D. $2^{(2n)} - 1$ by 3**
-

Q.4 Algorithm is a more general term in that the term ---- refers to a particular programming language.

- A. None
 - B. Information
 - C. Program**
 - D. Task
-

Q.5 The rational number '2' is the product of which of the two irrational numbers.

- A. $\sqrt{2}, \sqrt{7}$
 - B. $\sqrt{2}, \sqrt{3}$
 - C. $\sqrt{2}, \sqrt{2}$**
 - D. $\sqrt{2}, \sqrt{5}$
-

Q.6 'Reductio ad absurdum' is another name of -----

- A. Direct Method of proof
 - B. proof by contrapositive
 - C. None of these.
 - D. proof by contradiction**
-

Q.7 The sum of a rational and irrational number

- A. is always an irrational number**
 - B. is always a rational number
 - C. none of these
 - D. may or may not be rational number
-

Q.8 Set of prime numbers is finite.

- A. True
 - B. False**
-

Q.9 Proof byuses the equivalence $p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$

- A. superposition
 - B. mathematical induction
 - C. contradiction**
 - D. contraposition
-

Q.1 The loop is correct if the four properties are true, one of the property is?

A. Eventual Falsity of Guard

- B. Transitivity
- C. Commutativity
- D. Associativity

Q.2 Which of the following is true for $n = 2$?

A. $x^n - y^n$ is divisible by $x - y$.

- B. $x^n - y^n$ is divisible by $(x - y)^2$.
- C. $x^n - y^n$ is divisible by $x^3 - y^3$.
- D. $x^n - y^n$ is divisible by $x^5 - y^5$.

Q.3 The first step of describing algorithms formally is ?

A. The name of the algorithm, together with a list of input and output variables.

- B. An end statement.
- C. The output variable names, labeled by data type.
- D. The statements that make the body of the algorithm.

Q.4 The sum of two integers is even when

- A. one is prime and one is composite
- B. one is even and one is odd
- C. one is zero and one is odd
- D. both are even or both are odd**

Q.5 The predicate describing the initial state is

called None

B. pre-condition of the algorithm

- C. post-condition of the algorithm
- D. Out put variables

Q.6 A predicate becomes ----- when its variables are given specific values.

A. statement

- B. iteration
- C. algorithm
- D. sentence

Q.7 The sum of two odd integers is

- A. odd
- B. even**

Q.8 The sum of two irrational numbers must be an irrational number.

- A. True
 - B. False**
-

Q.9 The sum of two integers is not an even integer when

- A. one is even and one is odd**
 - B. both are even
 - C. both are odd
 - D. one is 0 and one is even
-

Q.10 A predicate is a sentence that contains a finite number of -----

- A. variables**
 - B. Constants
-

Q.1 Algorithm is a more general term in that the term ---- refers to a particular programming language.

- A. Program**
 - B. Information
 - C. None
 - D. Task
-

Q.2 Proof byuses the equivalence $p \rightarrow q \equiv \sim q \rightarrow \sim p$.

- A. contraposition**
 - B. mathematical induction
 - C. superposition
 - D. contradiction
-

Q.3 An integer 'n' is if it can be represented as a multiple of 2.

- A. prime
 - B. odd
 - C. composite
 - D. even**
-

Q.4 The quantity $a - d \cdot q$ equals the?

- A. Divisor
 - B. Remainder**
 - C. Quotient
 - D. None
-

Q.5 A predicate is a sentence that contains a finite number of -----

- A. variables**
 - B. Constants
-

Q.6 The product of any two consecutive positive integers is divisible by 2.

- A. True**
 - B. False
-

**Q.7 For every prime number n , $n + 2$ is prime.
statement.**

Which of the following prime number disproves the above

- A. $n=7$**
 - B. $n=5$
 - C. $n=11$
 - D. $n=3$
-

Q.8 A proof by contraposition is based on the logical equivalence between a statement and its.....

A. contrapositive

B. inverse

C. none of these

D. converse

Q.9 If n is an odd integer then ' $3n+2$ ' is

A. either even or odd

B. odd

C. neither even nor odd

D. even

Q.10 Set of prime numbers is finite.

A. True

B. False

Q.1 $(-1)^n = 1$ for $n = \dots$

(^ stands for power)

B. 7

C. 4

D. 3

Q.2 The sum of two odd integers is

A. even

B. odd

Q.3 For integers a, b, c , if a divides b and a divides c , then a divides $(a + b)$.

A. False

B. True

Q.4 In the Direct Proof, we show that if the statement p is true, then the statement q is -----

A. true

B. not necessarily true.

C. opposite to p

D. false

Q.5 In inductive property the first iteration of the loop is solved for?

A. $I(k + 1)$

B. None

C. $I(k)$

D. $I(0)$

Q.6 The first step of describing algorithms formally is ?

A. An end statement.

B. The statements that make the body of the algorithm.

C. The output variable names, labeled by data type.

D. The name of the algorithm, together with a list of input and output variables.

Q.7 The sum of two integers is not an even integer when

A. both are even

B. one is 0 and one is even

C. one is even and one is odd

D. both are odd

Q.8 The Product of an even and odd integer is

A. even

B. odd

Q.9 The sum of a rational and irrational number

A. may or may not be rational number

B. none of these

C. is always a rational number

D. is always an irrational number

Q.10 For all positive real numbers a and b, if $a < b$, then

A. $a^2 > b^2$

B. $a^2 = b^2$

C. None of these.

D. $a^2 < b^2$

Q.1 The sum of two irrational numbers in general

- A. is always a rational number
 - B. is always an irrational number
 - C. may or may not be an irrational number**
 - D. none of these
-

Q.2 If 'n' is an odd integer then n^3+n is

- A. odd
 - B. even**
-

Q.3 The Product of an even and odd integer is

- A. even**
 - B. odd
-

Q.4 The sum of two integers is not an even integer when

- A. both are even
 - B. one is 0 and one is even
 - C. one is even and one is odd**
 - D. both are odd
-

Q.5 Which of the following is not an irrational number?

- A. $\sqrt{4}$**
 - B. $\sqrt{5}$
 - C. $\sqrt{2}$
 - D. $\sqrt{3}$
-

Q.6 For every prime number n, $n + 2$ is prime. statement.

Which of the following prime number disproves the above

- A. $n=3$
 - B. $n=5$
 - C. $n=11$
 - D. $n=7$**
-

Q.7 The sum of two irrational numbers must be an irrational number.

- A. False**
 - B. True
-

Q.8 The quantity $a - d \cdot q$ equals the?

A. Remainder

B. Quotient

C. Divisor

D. None

Q.9 The sum of a rational and irrational number

A. is always an irrational number

B. may or may not be rational number

C. none of these

D. is always a rational number

Q.10 The loop is correct if the four properties are true, one of the property is?

A. Associativity

B. Transitivity

C. Eventual Falsity of Guard

D. Commutativity

Q.1 GCD of (9, 27) will be ____.

- A. 3
 - B. 9**
 - C. 5
 - D. 27
-

Q.2 Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

- A. 40**
 - B. 8
 - C. 5
 - D. 13
-

Q.3 A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement ?

- A. 8
 - B. 243
 - C. 15
 - D. 125**
-

Q.4 ----- is the mathematics of counting and arranging objects?

- A. Combinatorics**
 - B. Algebra
-

Q.5 If $k = 10$ and $n = 2$; find the number of ways using k-Selection.

- A. 11
 - B. 1
 - C. None of these**
 - D. 10
-

Q.6 The method of loop invariants is used to prove ----- of a loop with respect to certain pre and post-conditions.

- A. falseness
 - B. correctness**
-

Q.7 GCD of (8, 12) will be ____.

- A. 4**
 - B. 6
 - C. 3
 - D. 5
-

Q.8 In a city, the bus route numbers consist of a natural number less than 20, followed by one of the letters A,B,C,D,E and F. How many different bus routes are possible?

- A. 26
 - B. 114**
 - C. 120
 - D. 20!
-

Q.9 The greatest common divisor (gcd) of two integers a and b is the largest integer that divides both a and b. It is called -----

- A. Pre-Condition of Algorithm
 - B. the Euclidean Algorithm**
 - C. the Division Algorithm
 - D. Post-Condition of Algorithm
-

Q.10 There are three bus lines between A and B, and two bus lines between B and C. a person can travel by bus from A to C by way of B ? Find the number of ways

- A. 6**
 - B. 11
 - C. 13
 - D. 5
-

Q.1 $C(n,1)=\dots\dots\dots$

- A. 0
 - B. n**
 - C. 1
 - D. Cannot be determined
-

Q.2 In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

- A. 240
 - B. 10**
 - C. 1440
 - D. 30
-

Q.3 There are 4 bus lines between X and Y; and 5 bus lines between Y and Z, the number of ways a person can travel from X to Z?

- A. 20**
 - B. 9
-

Q.4 Number of ways the '7' student can be divided into group of '3' and '4' are

- A. 50
 - B. 120
 - C. 210
 - D. 35**
-

Q.5 GCD of (330,156) is?

- A. 8
 - B. 5
 - C. 12
 - D. 6**
-

Q.6 ----- is used to determine number of ordered or unordered arrangement of objects.

- A. Algorithm
 - B. Counting**
-

Q.7 $P(0, 0)=\underline{\hspace{1cm}}$?

- A. 0
 - B. Undefined
 - C. 2
 - D. 1**
-

Q.8 The method of loop invariants is used to prove ----- of a loop with respect to certain pre and post-conditions.

A. falseness

B. correctness

Q.9 If one event can occur in $\{n_1\}$ ways, a second event can occur in $\{n_2\}$ ways, a third event can occur in $\{n_3\}$ ways, how ways in which exactly one of the events can occur?

A. $\{n_1\} \cdot \{n_2\} \cdot \{n_3\}$

B. $\{n_1\} + \{n_2\} + \{n_3\}$

Q.10 $C(n, k) = k! P(n, k)$.

A. False

B. True

Q.1 Let $X = \{1, 2, 3\}$, then 2-combinations of the 3 elements of the set X are _____ ?

- A. $\{1, 2\}$, $\{2, 1\}$, $\{1, 3\}$ and $\{3, 1\}$
 - B. $\{1, 2\}$, $\{1, 3\}$ and $\{2, 3\}$**
 - C. $\{1, 2\}$, $\{2, 1\}$, $\{1, 3\}$ and $\{2, 3\}$
 - D. $\{1, 2\}$, $\{2, 1\}$, $\{1, 3\}$, $\{3, 1\}$, $\{2, 3\}$, and $\{3, 2\}$
-

Q.2 There are 5 Chinese books and 6 English books, a Student wants to select one optional book for both subject, total number of choices will be?

- A. 11
 - B. 10
 - C. 15
 - D. 30**
-

Q.3 A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions.

- A. 316
 - B. 216
 - C. None of these
 - D. 126**
-

Q.4 Sum rule in terms of sets, involves which set operation?

- A. Intersection
 - B. Union**
-

Q.5 $0! = \text{-----}$

- A. 1**
 - B. 0
 - C. -1
 - D. None of these
-

Q.6 GCD of (330,156) is?

- A. 6**
 - B. 8
 - C. 12
 - D. 5
-

Q.7 In how many ways a student can choose one of each of the courses when he is offered 3 mathematics courses, 4 literature courses and 2 history courses.

- A. 14
 - B. 24**
 - C. 9
 - D. 288
-

Q.8 GCD of (9, 27) will be ____.

- A. 3
 - B. 5
 - C. 27
 - D. 9**
-

Q.9 $C(n,1)=\dots\dots\dots$

- A. 1
 - B. 0
 - C. Cannot be determined
 - D. n**
-

Q.10 If order matters and repetition is not allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

- A. K-Sample
 - B. K-Permutation**
 - C. K-Combination
 - D. K-Selection
-

Q.1 In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

- A. 30
- B. 1440
- C. 10**
- D. 240

Q.2 In how many ways can a set of six letters be selected from the English Alphabets?

- A. $C(6, 26)$
- B. None of these
- C. $C(26, 6)$**
- D. $C(14, 6)$

Q.3 ----- is the mathematics of counting and arranging objects?

- A. Combinatorics**
- B. Algebra

Q.4 $(-2)! = \underline{\hspace{1cm}}$?

- A. 2
- B. 0
- C. Undefined**
- D. -2

Q.5 A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively.

How many possible projects are there to choose from ?

- B. 30**
- C. 216
- D. 18

Q.6 A boy can choose from 5 train services and 3 bus services to go to his hometown, How many total options he can choose from?

- A. 8
- B. 15**

Q.7 There are three bus lines between A and B, and two bus lines between B and C. person can travel round trip by bus from A to C by way of B ? Find the number of ways a

- A. 5
- B. 10
- C. 36**
- D. 6

Q.8 Number of ways the '7' student can be divided into group of '3' and '4' are

- A. 35
 - B. 210
 - C. 120
 - D. 50
-

Q.9 Let $X = \{1, 2, a, b\}$. Then 3-combinations of the 4 elements of the set X are _____ ?

- A. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{1, a, b\}$, and $\{b, a, 1\}$
 - B. $\{1, 2, a\}$, $\{1, 2, b\}$, $\{1, a, b\}$ and $\{2, a, b\}$
 - C. $\{1, b, a\}$, $\{a, b, 2\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
 - D. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
-

Q.10 Which of the followings is correct option?

- A. $C(n, k) = C(n + k, n - k)$
 - B. $C(n, n) = n$
 - C. $C(n, k) + C(n, k + 1) = C(n, k + 1)$
 - D. $C(n, 0) = 1$
-

Q.1 If one event can occur in $\{n_1\}$ ways, a second event can occur in $\{n_2\}$ ways, a third event can occur in $\{n_3\}$ ways, how ways all of the events can occur in the order?

- A. $\{n_1\}.\{n_2\}.\{n_3\}$
 - B. $\{n_1\} + \{n_2\} + \{n_3\}$**
-

Q.2 How many multiples of 5 are there from 10 to 75?

- A. 15
 - B. 20
 - C. 10
 - D. 14**
-

Q.3 The method of loop invariants is used to prove ----- of a loop with respect to certain pre and post-conditions.

- A. falseness
 - B. correctness**
-

Q.4 How many possible outcomes are there when a fair coin is tossed four times ?

- A. 4
 - B. 16**
 - C. 32
 - D. 8
-

Q.5 A boy can choose from 5 train services and 3 bus services to go to his hometown, How many total options he can choose from?

- A. 8
 - B. 15**
-

Q.6 $(n-2)!/(n-2)=$ _____ ?

- A. $(n-2)!$
 - B. $(n-2)(n-3)!$
 - C. $(n-3)!$**
 - D. $(n-1)!$
-

Q.7 If one event can occur in $\{n_1\}$ ways, a second event can occur in $\{n_2\}$ ways, a third event can occur in $\{n_3\}$ ways, how ways in which exactly one of the events can occur?

- A. $\{n_1\} + \{n_2\} + \{n_3\}$**
 - B. $\{n_1\}.\{n_2\}.\{n_3\}$
-

Q.8 $P(2,2)=\dots\dots\dots$

- A. 0
 - B. 1
 - C. 4!
 - D. 2!**
-

Q.9 If order matters and repetition is not allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

- A. K-Permutation**
 - B. K-Selection
 - C. K-Sample
 - D. K-Combination
-

Q.10 There are 4 bus lines between X and Y; and 5 bus lines between Y and Z , the number of ways a person can travel from X to Z?

- A. 20**
 - B. 9
-

Q.1 Which of the followings is the factorial form of $5 \cdot 4$?

- A. $5!/3$
- B. $5/3$
- C. $5/3!$
- D. $5!/3!$**

Q.2 If order matters and repetition is allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

- A. K-Sample
- B. K-Permuatation**
- C. K-combination
- D. K-Selection

Q.3 GCD of (8, 12) will be ____.

- A. 4**
- B. 6
- C. 5
- D. 3

Q.4 The chairs of an auditorium are to be labeled with two characters, a letter followed by a digit. Letters are a and b and numbers are 3 and 5. What is the largest number of chairs that can be labeled differently ?

- A. 10
- B. 8
- C. 4**
- D. 15

Q.5 A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions.

- A. 316
- B. 126**
- C. None of these
- D. 216

Q.6 A bit string is a sequence of 0's and 1's.

- A. False
- B. True**

Q.7 Which of the followings is correct option?

- A. $C(n, n) = n$
 - B. $C(n, k) + C(n, k + 1) = C(n, k + 1)$**
 - C. $C(n, 0) = 1$
 - D. $C(n, k) = C(n + k, n - k)$
-

Q.8 A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement ?

- A. 15
 - B. 8
 - C. 243
 - D. 125**
-

Q.9 How many multiples of 5 are there from 10 to 75?

- A. 10
 - B. 20
 - C. 15
 - D. 14**
-

Q.10 $P(2,2)=\dots\dots\dots$

- A. 4!
 - B. 0
 - C. 1
 - D. 2!**
-

Q.1 $(-2)! = \underline{\hspace{1cm}}$?

A. Undefined

B. -2

C. 0

D. 2

Q.2 Number of ways the '7' student can be divided into group of '3' and '4' are

A. 35

B. 210

C. 120

D. 50

Q.3 Which of the followings is the factorial form of $5 \cdot 4$?

A. $5/3$

B. $5!/3!$

C. $5/3!$

D. $5!/3$

Q.4 Sum rule in terms of sets, involves which set operation?

A. Intersection

B. Union

Q.5 In how many ways can a set of six letters be selected from the English Alphabets?

A. $C(26, 6)$

B. $C(14, 6)$

C. $C(6, 26)$

D. None of these

Q.6 A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions.

A. 216

B. 126

C. 316

D. None of these

Q.7 If order matters and repetition is not allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

A. K-Sample

B. K-Selection

C. K-Combination

D. K-Permutation

Q.8 Compute $C(8, 3)$

- A. 46
 - B. 36
 - C. 56**
 - D. 65
-

Q.9 Which of the followings is correct option?

- A. $C(n, n) = n$
 - B. $C(n, 0) = n$
 - C. $C(n, 1) = n$**
 - D. $C(n, k) = C(n + k, n - k)$
-

Q.10 Combination is used when we have to select 'k' elements from a set of 'n' elements with the following properties.

- A. Order matters and repetition is not allowed.
 - B. Order does not matter and repetition is allowed
 - C. Order does not matter and repetition is not allowed**
 - D. Order matters and repetition is allowed
-

Q.1 Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

- A. 8
- B. 13
- C. 40**
- D. 5

Q.2 $P(0, 0) = \underline{\hspace{1cm}}$?

- A. Undefined
- B. 1
- C. 0**
- D. 2

Q.3 ----- is used to determine number of ordered or unordered arrangement of objects.

- A. Counting**
- B. Algorithm

Q.4 If one event can occur in $\{n_1\}$ ways, a second event can occur in $\{n_2\}$ ways, a third event can occur in $\{n_3\}$ ways, how ways in which exactly one of the events can occur?

- A. $\{n_1\} + \{n_2\} + \{n_3\}$
- B. $\{n_1\} \cdot \{n_2\} \cdot \{n_3\}$**

Q.5 Let $X = \{1, 2, a, b\}$. Then 3-combinations of the 4 elements of the set X are _____ ?

- A. $\{1, b, a\}$, $\{a, b, 2\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
- B. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{1, a, b\}$, and $\{b, a, 1\}$
- C. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
- D. $\{1, 2, a\}$, $\{1, 2, b\}$, $\{1, a, b\}$ and $\{2, a, b\}$**

Q.6 $0! = \text{-----}$

- A. 1**
- B. None of these
- C. 0
- D. -1

Q.7 The method of loop invariants is used to prove ----- of a loop with respect to certain pre and post-conditions.

- A. correctness**
- B. falseness

Q.8 GCD of (9, 27) will be ____.

- A. 3
 - B. 9**
 - C. 5
 - D. 27
-

Q.1 The greatest common divisor (gcd) of two integers a and b is the largest integer that divides both a and b. It is called -----

- A. the Division Algorithm
- B. Post-Condition of Algorithm
- C. Pre-Condition of Algorithm
- D. the Euclidean Algorithm**

Q.2 There are three bus lines between A and B, and two bus lines between B and C. person can travel round trip by bus from A to C by way of B ? Find the number of ways a

- A. 10
- B. 5
- C. 6
- D. 36**

Q.3 How many multiples of 5 are there from 10 to 75?

- A. 20
- B. 10
- C. 14**
- D. 15

Q.4 $P(0, 0) =$ _____ ?

- A. 1
- B. 2
- C. 0**
- D. Undefined

Q.5 $(n-2)!/(n-2) =$ _____ ?

- A. $(n-2)!$
- B. $(n-1)!$
- C. $(n-3)!$**
- D. $(n-2)(n-3)!$

Q.6 A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively.

How many possible projects are there to choose from ?

- A. 216**
- B. 216
- C. 12
- D. 18

Q.7 In a city, the bus route numbers consist of a natural number less than 20, followed by one of the letters A,B,C,D,E and F. How many different bus routes are possible?

- A. 120
 - B. 20!
 - C. 26
 - D. 114**
-

Q.8 Which of the followings is the factorial form of $5 \cdot 4$?

- A. $5!/3!$**
 - B. $5/3!$
 - C. $5!/3$
 - D. $5/3$
-

Q.9 Which of the followings is correct option?

- A. $C(n, 0) = 1$**
 - B. $C(n, k) + C(n, k + 1) = C(n, k + 1)$
 - C. $C(n, n) = n$
 - D. $C(n, k) = C(n + k, n - k)$
-

Q.10 $C(n, k) = k! P(n, k)$.

- A. False**
 - B. True
-

Q.1 GCD of (8, 12) will be ____.

- A. 5
 - B. 4**
 - C. 6
 - D. 3
-

Q.2 Sum rule in terms of sets, involves which set operation?

- A. Union**
 - B. Intersection
-

Q.3 In how many ways can 6 people be seated on 6 available seats?

- A. 120
 - B. 720**
 - C. 6
 - D. 12
-

Q.4 How many multiples of 5 are there from 10 to 75?

- A. 15
 - B. 20
 - C. 10
 - D. 14**
-

Q.5 If there are 5 different optional courses in English and 3 student who wants to take one optional course ____? different optional courses in Maths. Choices for a

- A. 9**
 - B. 5
 - C. 8
 - D. 7
-

Q.6 Which of the followings is correct option?

- A. $C(n, k) + C(n, k + 1) = C(n, k + 1)$
 - B. $C(n, n) = n$
 - C. $C(n, k) = C(n + k, n - k)$
 - D. $C(n, 0) = 1$**
-

Q.7 $(n-2)!/(n-2)=$ _____ ?

- A. $(n-3)!$**
 - B. $(n-2)!$
 - C. $(n-2)(n-3)!$
 - D. $(n-1)!$
-

Q.8 Let $X = \{1, 2, a, b\}$. Then 3-combinations of the 4 elements of the set X are _____ ?

- A. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{1, a, b\}$, and $\{b, a, 1\}$
 - B. $\{1, 2, a\}$, $\{1, 2, b\}$, $\{1, a, b\}$ and $\{2, a, b\}$**
 - C. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
 - D. $\{1, b, a\}$, $\{a, b, 2\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
-

Q.9 The value of $(-3)!$ = _____.

- A. -6
 - B. Not defined**
 - C. 6
 - D. None of the above
-

Q.1 ----- is the mathematics of counting and arranging objects?

A. Combinatorics

B. Algebra

Q.2 Find the value of $C(n, n)$

A. n

B. None of these

C. 1

D. 0

Q.3 ----- is used to determine number of ordered or unordered arrangement of objects.

A. Counting

B. Algorithm

Q.4 A bit string is a sequence of 0's and 1's.

A. True

B. False

Q.5 Let $X = \{1, 2, a, b\}$. Then 3-combinations of the 4 elements of the set X are _____ ?

A. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$

B. $\{1, b, a\}$, $\{a, b, 2\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$

C. $\{1, 2, a\}$, $\{1, 2, b\}$, $\{1, a, b\}$ and $\{2, a, b\}$

D. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{1, a, b\}$, and $\{b, a, 1\}$

Q.6 How many possible outcomes are there when a fair coin is tossed four times ?

A. 8

B. 16

C. 32

D. 4

Q.7 If order matters and repetition is allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

A. K-Sample

B. K-Selection

C. K-Permutation

D. K-combination

Q.8 The value of $(-3)!$ = ____.

- A. 6
 - B. Not defined**
 - C. -6
 - D. None of the above
-

Q.9 In how many ways can 6 people be seated on 6 available seats?

- A. 720**
 - B. 12
 - C. 120
 - D. 6
-

Q.10 GCD of (8, 12) will be ____.

- A. 4**
 - B. 3
 - C. 5
 - D. 6
-

Q.1 How many multiples of 5 are there from 10 to 75?

- A. 15
 - B. 20
 - C. 10
 - D. 14**
-

Q.2 How many possible outcomes are there when a fair coin is tossed four times ?

- A. 32
 - B. 4
 - C. 16**
 - D. 8
-

Q.3 There are 4 bus lines between X and Y; and 5 bus lines between Y and Z , the number of ways a person can travel from X to Z?

- A. 20**
 - B. 9
-

Q.4 How many 4-bit string contain at least one "1"

- A. 16
 - B. 12
 - C. 15**
 - D. 14
-

Q.5 Let $X = \{1, 2, 3\}$, then 2-combinations of the 3 elements of the set X are _____ ?

- A. $\{1, 2\}$, $\{1, 3\}$ and $\{2, 3\}$**
 - B. $\{1, 2\}$, $\{2, 1\}$, $\{1, 3\}$ and $\{2, 3\}$
 - C. $\{1, 2\}$, $\{2, 1\}$, $\{1, 3\}$ and $\{3, 1\}$
 - D. $\{1, 2\}$, $\{2, 1\}$, $\{1, 3\}$, $\{3, 1\}$, $\{2, 3\}$, and $\{3, 2\}$
-

Q.6 GCD of (9, 27) will be _____.

- A. 9**
 - B. 3
 - C. 27
 - D. 5
-

Q.7 ----- is used to determine number of ordered or unordered arrangement of objects.

- A. Algorithm
 - B. Counting**
-

Q.8 GCD of (8, 12) will be ____.

- A. 6
- B. 5
- C. 3
- D. 4

Q.9 A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively.

How many possible projects are there to choose from ?

- A. 30
- B. 12
- C. 18
- D. 216

Q.10 Which of the followings is the factorial form of $5 \cdot 4$?

- A. $5!/3$
- B. $5/3!$
- C. $5/3$
- D. $5!/3!$

Q.1 $0! = \text{-----}$

- A. None of these
 - B. 1**
 - C. 0
 - D. -1
-

Q.2 How many multiples of 5 are there from 10 to 75?

- A. 20
 - B. 15
 - C. 14**
 - D. 10
-

Q.3 A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement ?

- A. 8
 - B. 15
 - C. 125**
 - D. 243
-

Q.4 Number of ways the '7' student can be divided into group of '3' and '4' are

- A. 210
 - B. 50
 - C. 35**
 - D. 120
-

Q.5 $C(n,0) = \text{-----}$

- A. Both b and c
 - B. 1**
 - C. n
 - D. $C(n,n)$
-

Q.6 Combination is used when we have to select 'k' elements from a set of 'n' elements with the following properties.

- A. Order matters and repetition is not allowed.
 - B. Order matters and repetition is allowed
 - C. Order does not matter and repetition is allowed
 - D. Order does not matter and repetition is not allowed**
-

Q.7 The method of loop invariants is used to prove ----- of a loop with respect to certain pre and post-conditions.

A. correctness

B. falseness

Q.8 $(-2)! = \underline{\hspace{2cm}}$?

A. 2

B. -2

C. 0

D. Undefined

Q.9 There are three bus lines between A and B, and two bus lines between B and C. person can travel round trip by bus from A to C by way of B ?

Find the number of ways a

A. 10

B. 5

C. 6

D. 36

Q.10 In how many ways a student can choose one of each of the courses when he is offered 3 mathematics courses, 4 literature courses and 2 history courses.

A. 14

B. 288

C. 9

D. 24

Q.1 Compute $C(8, 3)$

A. 56

B. 36

C. 46

D. 65

Q.2 The value of $(-3)!$ = ____.

A. None of the above

B. 6

C. Not defined

D. -6

Q.3 Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

A. 8

B. 13

C. 5

D. 40

Q.4 Number of combinations of 3-bit string that contain exactly three "0" is

A. 8

B. 7

C. 1

D. 6

Q.5 A bit string is a sequence of 0's and 1's.

A. True

B. False

Q.6 If there are 5 different optional courses in English and 3 student who wants to take one optional course ____? different optional courses in Maths. Choices for a

A. 9

B. 7

C. 5

D. 8

Q.7 A boy can choose from 5 train services and 3 bus services to go to his hometown, How many total options he can choose from?

A. 15

B. 8

Q.8 The chairs of an auditorium are to be labeled with two characters, a letter followed by a digit. Letters are a and b and numbers are 3 and 5. What is the largest number of chairs that can be labeled differently ?

- A. 4**
 - B. 8
 - C. 15
 - D. 10
-

Q.9 $(-2)! = \underline{\hspace{1cm}} ?$

- A. 0
 - B. 2
 - C. Undefined**
 - D. -2
-

Q.10 A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions.

- A. 216
 - B. None of these
 - C. 316
 - D. 126**
-

Q.1 Which of the followings is correct option?

- A. $C(n, k) = C(n + k, n - k)$
- B. $C(n, k) + C(n, k + 1) = C(n, k + 1)$
- C. $C(n, 0) = 1$**
- D. $C(n, n) = n$

Q.2 How many possible outcomes are there when a fair coin is tossed four times ?

- A. 4
- B. 32
- C. 16**
- D. 8

Q.3 If one event can occur in $\{n_1\}$ ways, a second event can occur in $\{n_2\}$ ways, a third event can occur in $\{n_3\}$ ways, how ways all of the events can occur in the order?

- A. $\{n_1\} \cdot \{n_2\} \cdot \{n_3\}$**
- B. $\{n_1\} + \{n_2\} + \{n_3\}$

Q.4 There are three bus lines between A and B, and two bus lines between B and C. person can travel round trip by bus from A to C by way of B ? Find the number of ways a

- A. 6
- B. 36**
- C. 10
- D. 5

Q.5 The chairs of an auditorium are to be labeled with two characters, a letter followed by a digit. Letters are a and b and numbers are 3 and 5. What is the largest number of chairs that can be labeled differently ?

- A. 8
- B. 15
- C. 10
- D. 4**

Q.6 which of the followings is the correct option for $9!/0!$?

- A. 9
- B. 1
- C. 0
- D. 9!**

Q.7 In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

- A. 30
 - B. 1440
 - C. 10**
 - D. 240
-

Q.8 Let $X = \{1, 2, a, b\}$. Then 3-combinations of the 4 elements of the set X are _____ ?

- A. $\{1, b, a\}$, $\{a, b, 2\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
 - B. $\{1, 2, a\}$, $\{1, 2, b\}$, $\{1, a, b\}$ and $\{2, a, b\}$**
 - C. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{b, 1, a\}$ and $\{1, 2, b\}$
 - D. $\{1, 2, a\}$, $\{a, 2, 1\}$, $\{1, a, b\}$, and $\{b, a, 1\}$
-

Q.9 $0! = \text{-----}$

- A. -1
 - B. 0
 - C. None of these
 - D. 1**
-

Q.10 There are 4 bus lines between X and Y; and 5 bus lines between Y and Z, the number of ways a person can travel from X to Z?

- A. 20**
 - B. 9
-